

Foundry Tools Cheat Sheet

Baseline 2026-10-06: Azure CLI 2.90.0, Bicep 0.47.16. Foundry Tools = formerly Azure AI services / Cognitive Services. Bookshelf: \$RG=rg-bookshelf, \$LOCATION=eastus2, \$FOUNDRY=foundry-bookshelf, project proj-bookshelf, search-bookshelf. Quick reference only — verify against the official docs.

CATALOG AND USE-CASE FINDER				
Tool	Status	Use it to...	REST path (api-version)	Python / .NET package
Speech STT	GA; LLM speech preview	transcribe calls, podcasts, captions	/speechtotext/transcriptions:transcribe, :submit (2025-10-15)	azure-cognitiveservices-speech / Microsoft.CognitiveServices.Speech 1.52
Speech TTS	GA	narration, read-aloud, IVR, avatars	/texttospeech/batchsyntheses/{id} (2024-04-01)	same Speech SDK 1.52
Voice Live	GA	voice agents (speech to speech)	wss://<res>.services.ai.azure.com/voice-live/realtime (2026-04-10)	azure-ai-voicelive 1.3 / Azure.AI.VoiceLive 1.2
Translator text	GA	translate strings, NMT or LLM per target	/translate (2026-06-06; v3.0 still GA)	azure-ai-translation-text 2.0 / Azure.AI.Translation.Text 2.0
Translator documents	GA	translate DOCX/PDF keeping layout	/translator/document/batches (2026-03-01)	azure-ai-translation-document 2.0 / .Document 3.0
Language core	GA	redact PII, NER, detect language, health	/language/:analyze-text (2024-11-01; PII 2026-05-01)	azure-ai-textanalytics 5.4 / Azure.AI.TextAnalytics 5.3
Language legacy	ends 2029-03-31	sentiment, summaries, CLU, CQA	/language/:analyze-conversations (2024-11-01)	same; move to Foundry Models
Vision Image Analysis 4.0	deprecated 2028-09-25	captions, tags, smart crops, OCR	/computervision/imageanalysis:analyze (2024-02-01)	azure-ai-vision-imageanalysis 1.0 / Azure.AI.Vision.ImageAnalysis 1.0
Face	Limited Access; SDK preview	detect, verify, liveness (KYC)	/face/v1.2/detect, /detectLivenessWithVerify-sessions	azure-ai-vision-face 1.0.0b2 / Azure.AI.Vision.Face beta
Document Intelligence	GA	OCR, layout, invoices, receipts, IDs	/documentintelligence/documentModels/{id}:analyze (2024-11-30)	azure-ai-documentintelligence 1.0.2 / Azure.AI.DocumentIntelligence 1.0
Content Understanding	GA; agentic preview	fields you describe, video chapters	/contentunderstanding/analyzers/{id}:analyze (2025-11-01)	azure-ai-contentunderstanding 1.1 / Azure.AI.ContentUnderstanding 1.1
Content Safety	GA; some preview	moderate reviews, block jailbreaks	/contentsafety/text:analyze, text:shieldPrompt (2024-09-01)	azure-ai-contentsafety 1.0 / Azure.AI.ContentSafety 1.0
AI Search	GA (own resource)	hybrid search, enrichment, agentic retrieval	<svc>.search.windows.net/indexes, /knowledgebases/{kb}/retrieve (2026-04-01)	azure-search-documents 12.0 / Azure.Search.Documents 12.0
Immersive Reader	GA (own kind)	read-aloud and reading aids in a page	JS SDK launchAsync(token, subdomain, content)	@microsoft/immersive-reader-sdk

RESOURCE KINDS

- AIServices** = Foundry resource: all tools + Foundry Models + projects. Default choice.
- CognitiveServices**: legacy multi-service; convert with account update --kind AIServices.
- Single-service**: FormRecognizer, Face, TextAnalytics (+ SpeechServices, TextTranslation, ComputerVision, ContentSafety, ImmersiveReader; check list-kinds). Needed for commitment tiers, disconnected containers.
- Custom subdomain: required for Entra ID and VNets; **cannot be changed**.

ENDPOINTS

- Tools: https://<name>.cognitiveservices.azure.com/
- Project: https://<name>.services.ai.azure.com/api/projects/<proj>
- Models: https://<name>.openai.azure.com/
- Speech legacy: <region>.stt|tts.speech.microsoft.com
- Translator global: api.cognitive.microsofttranslator.com + Ocp-Apim-Subscription-Region

AUTH HEADERS AND ROLES

- Key: Ocp-Apim-Subscription-Key (KEY1/KEY2, rotate one at a time).
- STS: POST /sts/v1.0/issueToken → 10-min token (Speech, Translator).
- Entra: Authorization: Bearer, scope cognitiveservices.azure.com/.default; DefaultAzureCredential.
- Roles: **Cognitive Services User**; Speech User, Language Reader, Face Recognizer; Foundry (Azure AI) User / Project Manager / Account Owner / Owner.
- Owner/Contributor: no data actions (401).
- Keys off: disableLocalAuth=true + Azure Policy.

KEY CLI COMMANDS

```
az cognitiveservices account create \
-n $FOUNDRY -g $RG -l $LOCATION \
--kind AIServices --sku S0 \
--custom-domain $FOUNDRY \
--assign-identity --yes \
--allow-project-management
az cognitiveservices account \
project create -n $FOUNDRY -g $RG \
--project-name proj-bookshelf
az cognitiveservices account \
keys list -n $FOUNDRY -g $RG
az cognitiveservices account \
list-usage -n $FOUNDRY -g $RG
az cognitiveservices account list-kinds
az role assignment create \
--assignee <principal-id> \
--role "Cognitive Services User" \
--scope <foundry-resource-id>
az cognitiveservices account \
deployment create \
-n $FOUNDRY -g $RG \
--deployment-name gpt-4.1-mini \
--model-name gpt-4.1-mini \
--model-format OpenAI \
--model-version 2025-04-14 \
--sku-name GlobalStandard
```

CHOOSING

- Prebuilt tool**: narrow, high volume, low latency, regulated, offline (containers).
- Foundry Model**: judgment, open output, multi-step; per-token cost.
- Custom model**: prebuilt close but not accurate enough; labeled data.
- OCR: Document Intelligence Read (not Image Analysis).
- Documents: DI for known forms; Content Understanding for described fields.

SPEECH SNIPPETS

```
cfg = SpeechConfig(endpoint=EP,
token_credential=
DefaultAzureCredential())
curl -F audio=@a.mp3 \
-F 'definition={"locales":["en-US"],
"diarization":{"enabled":true}}' \
"$EP/speechtotext/transcriptions\
:transcribe?api-version=2025-10-15"
```

- Fast: 500 MB / 5 h, 35 speakers. Batch: Blob in/out, polling. TTS real-time max 10 min; longer = batch synthesis.

TRANSLATOR 2026-06-06

```
{"inputs":[{"text":"...",
"targets":[
{"language":"es"},
{"language":"fr"},
{"deploymentName":"gpt-4.1-mini",
"tone":"formal"}]}}
```

- No deployment = NMT (per char). With deployment = LLM (per token, Foundry resource). No detect / breaksentence / dictionary: use Language.

LANGUAGE PII (2026-05-01)

```
{"kind":"PiiEntityRecognition",
"parameters":{"redactionPolicies":[{"policyKind":
"syntheticReplacement"}],
"excludedValues":["Bookshelf"]},
"analysisInput":{...}}
```

- Sync: 5,120 chars/doc. Async jobs for health, custom NER, document PII.

DOCUMENTS

- DI async: 292 + Operation-Location, poll; results kept 24 h.
- Layout outputContentFormat=markdown for RAG chunks.
- Custom: template, neural, composed (200), classifier splitMode=auto.
- CU fields: extract / classify / generate; set default model deployments first.

SAFETY AND SEARCH

- Harm: hate, sexual, violence, self-harm; severity 0/2/4/6 (or 0–7).
- Prompt Shields: user prompt + documents (indirect injection).
- Groundedness (preview), protected material, task adherence.
- Prefer Foundry guardrails on model deployments.
- Search skillsets bill via attached Foundry resource (AIServicesByIdentity).
- Agentic retrieval: knowledge base + knowledge sources.

LIMITS HIGHLIGHTS

- Default ~10 TPS per tool; 429 + Retry-After; SDK retries; autoscale dynamicThrottlingEnabled.
- DI: 15 TPS, 500 MB / 2,000 pages (S0).
- Content Safety: 1,000 req / 10 s; 10k chars.
- Translator: 50,000 chars / request (NMT).
- Doc translation: 40 MB/doc, 1,000 files, 250 MB/batch.
- Language: 1,000 req/min.

NETWORK, CONTAINERS, MONITORING

- DNS zones: privatelink.cognitiveservices.azure.com, .openai.azure.com, .services.ai.azure.com.
- Containers: docker run ... Eula=accept Billing=<endpoint> ApiKey=<key>.
- Diagnostics: Audit, RequestResponse, AllMetrics → AzureDiagnostics.
- Limited Access: Face identify/verify, custom voice, custom avatar.

RETIREMENTS TIMELINE				
2025 : QnA Maker (03-31), background removal / Product Recognition (03-31), pronunciation content assessment (Jul), Speaker Recognition (09-30), intent recognition (Oct), Spatial Analysis	2026 : LUIS + STT REST v3.x (03-31); Anomaly Detector, Personalizer, Metrics Advisor (10-01); ConversationTranslator removed (SDK 1.51)	2027 : Content Moderator (03-15), Language Studio (03-20), classic agents (03-31), DI v2.1 (09-15)	2028 : entity linking (09-01), Image Analysis 3.2/4.0 + Custom Vision (09-25)	2029 : DI v3.0 (03-30), Language legacy features (03-31)